Assignment 9

Chapter 8

1. Obtain the forward channel description of $R(D)$ for the binary source with the Hamming distortion measure.

2. *Binary covering radius* The Hamming ball with center $\mathbf{c} = (c_1, c_2, \cdots, c_n) \in \{0,1\}^n$ and radius r is the set
$$S_r(\mathbf{c}) = \left\{ \mathbf{x} \in \{0,1\}^n : \sum_{i=1}^{n} |x_i - c_i| \le r \right\}.$$
Let $M_{r,n}$ be the minimum number M such that there exists Hamming balls $S_r(\mathbf{c}_j)$, $j = 1, 2, \cdots, M$ such that for all $\mathbf{x} \in \{0,1\}^n$, $\mathbf{x} \in S_r(\mathbf{c}_j)$ for some j.

   (a) Show that
$$M_{r,n} \ge \frac{2^n}{\sum_{k=0}^{r} \binom{n}{k}}.$$

   (b) What is the relation between $M_{r,n}$ and the rate-distortion function for the binary source with the Hamming distortion measure?

3. Consider a source random variable X with the Hamming distortion measure.

   (a) Prove that
$$R(D) \ge H(X) - D \log(|\mathcal{X}| - 1) - h_b(D)$$
   for $0 \le D \le D_{max}$.

   (b) Show that the above lower bound on $R(D)$ is tight if X is distributed uniformly on $\mathcal{X}$.

   See Jerohin [190] (also see [83], p.133) for the tightness of this lower bound for a general source. This bound is a special case of the Shannon lower bound for the rate-distortion function [327] (also see [79], p.369).

5. *Compound source* Let Θ be an index set and $\mathcal{Z}_\Theta = \{X_\theta : \theta \in \Theta\}$ be a collection of source random variables. The random variables in $\mathcal{Z}_\Theta$ have a common source alphabet $\mathcal{X}$, a common reproduction alphabet $\hat{\mathcal{X}}$, and a common distortion measure d. A compound source is an i.i.d. information source whose generic random variable is X_Φ, where Φ is equal to some $\theta \in \Theta$ but we do not know which one it is. The rate-distortion function $R_\Phi(D)$ for X_Φ has the same definition as the rate-distortion function defined in this chapter except that (8.23) is replaced by
$$\Pr\{d(\mathbf{X}_\theta, \hat{\mathbf{X}}) > D + \epsilon\} \le \epsilon \quad \text{for all } \theta \in \Theta.$$
   Show that
$$R_\Phi(D) = \sup_{\theta \in \Theta} R_\theta(D),$$
   where $R_\theta(D)$ is the rate-distortion function for X_θ.